Answer Key

Lesson Quizzes · Grade 9 Math · Term 1

1-1 · Numerical Expressions

1 C 2 A 3 D 4 B 5 A 6 B 7 A 8 B 9 A 10 B

11. (a) $2 \cdot 42 + 3 \cdot 34 + 2 \cdot 32 + 5 \cdot 4$ · (b) $84 + 102 + 64 + 20$ · = AED 270

1-2 · Algebraic Expressions

1 A 2 C 3 B 4 C 5 C 6 D 7 B 8 B 9 D 10 D

11. (a) $1.75 + 3.45m$ · (b) $1.75 + 3.45(8) = 1.75 + 27.60 = 29.35$ · AED 29.35

1-3 · Properties of Real Numbers

1 C 2 A 3 B 4 D 5 D 6 B 7 A 8 A 9 D 10 B

11. Commutative (×): $\frac{5}{9} \cdot \frac{9}{5} \cdot 9 \cdot 4.$ · Multiplicative Inverse: $1 \cdot 9 \cdot 4.$ · Multiplicative Identity: $9 \cdot 4 = 36.$

1-4 · Distributive Property

1 A 2 B 3 A 4 B 5 B 6 A 7 D 8 C 9 B 10 C

11. $3(2x + 3y) - 2(4x - y)$ · = $6x + 9y - 8x + 2y$ (Distributive) · = $6x - 8x + 9y + 2y$ (Commutative) · = $(6 - 8)x + (9 + 2)y = -2x + 11y$ (Distributive)

2-1 · Writing and Interpreting Equations

1 A 2 D 3 B 4 C 5 D 6 B 7 A 8 B 9 C 10 D

11. (a) $2c + 24 = 136$ · (b) $2c = 112$, $c = 56$ · 56 million packages of raisin bagels

2-2 · Solving One-Step Equations

1 D 2 D 3 D 4 C 5 D 6 B 7 C 8 C 9 A 10 C

11. (a) $\frac{2}{3}n = 96$ · (b) $n = 96 \cdot \frac{3}{2} = 144$ · (c) The whole class (144) must be more than the part (96).

2-3 · Solving Multi-Step Equations

1 C 2 C 3 A 4 A 5 D 6 B 7 D 8 A 9 C 10 B

11. (a) $\frac{2}{5}c + 150 = 630$ · (b) $\frac{2}{5}c = 480$, $c = 480 \cdot \frac{5}{2} = 1200$ · The trip costs AED 1200.

2-4 · Solving Equations with the Variable on Each Side

1 A 2 A 3 B 4 B 5 D 6 D 7 B 8 D 9 B 10 C

11. (a) $5(x + 4) = \frac{1}{2}(12)(2x - 6)$ · (b) $5x + 20 = 12x - 36 \cdot 56 = 7x$, $x = 8$

2-5 · Solving Equations Involving Absolute Value

1 B 2 C 3 D 4 B 5 D 6 B 7 D 8 A 9 B 10 D

11. (a) $|x - 2000| = 250$ · (b) $x - 2000 = 250$ → 2250; $x - 2000 = -250$ → 1750 · Maximum 2250, minimum 1750.

2-6 · Solving Proportions

1 A 2 D 3 D 4 B 5 A 6 C 7 B 8 B 9 C 10 B

11. (a) $\frac{140}{5} = \frac{c}{12}$ · (b) $c = \frac{12(140)}{5} = 336$ cm · (c) 28 cm per year × 12 = 336; and 10 years would be 280 cm, so a bit more makes sense.

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2-7 · Using Formulas

1 D 2 D 3 C 4 A 5 D 6 D 7 C 8 C 9 B 10 C

11. (a) $0.5 \cdot 4 \cdot 2 \cdot 2 \cdot 8 = 64$ fl oz · (b) $64 \div 20 = 3.2$ · She can make 3 full batches.

4-1 · Graphing Linear Functions

1 B 2 A 3 D 4 D 5 C 6 D 7 B 8 B 9 D 10 A

11. (a) y-int: 4362; x-int: $\frac{2}{3}x = 4362 \rightarrow x = 6543$ · (b) 4362 lb at the start; after 6543 cups all the peanuts are used.

4-2 · Rate of Change and Slope

1 C 2 D 3 C 4 B 5 C 6 B 7 B 8 A 9 B 10 C

11. (a) $\frac{1675 - 1350}{2005 - 2000} = \frac{325}{5} = 65$ · (b) The budget increased by AED 65 per year.

4-3 · Slope-Intercept Form

1 A 2 C 3 B 4 A 5 D 6 D 7 C 8 D 9 D 10 A

11. (a) $y = 0.06x + 4.61$ · (b) $x = 2$ means 2 months after May (July). · $y = 0.12 + 4.61 = 4.73$ million openings.

4-4 · Transformations of Linear Functions

1 B 2 D 3 C 4 A 5 D 6 B 7 A 8 D 9 B 10 C

11. (a) 2 units left and 4 units down. · (b) reflected across the x-axis and stretched vertically by 2. · (c) stretched horizontally (less steep).

4-5 · Arithmetic Sequences

1 B 2 B 3 D 4 C 5 B 6 B 7 D 8 A 9 A 10 C

11. (a) $d = 10.5$, $a_n = 10.5 + (n - 1)10.5 = 10.5n$ · (b) $10.5(26.2) = 275.1$ minutes

4-6 · Piecewise and Step Functions

1 C 2 C 3 C 4 A 5 B 6 B 7 B 8 C 9 B 10 D

11. (a) $0 < x \le 1$: 35 · $1 < x \le 2$: 70 · $2 < x \le 3$: 105 · $3 < x \le 4$: 140 · (b) 3.2 is in $3 < x \le 4$ → AED 140

4-7 · Absolute Value Functions

1 D 2 D 3 D 4 B 5 C 6 A 7 A 8 D 9 A 10 A

11. (a) reflected across the x-axis, 3 units left, 5 units up · (b) vertex (−3, 5) · (c) $y \le 5$